Geo-SimCLR: Self-Supervised Contrastive Pre-training of Sentinel-2 for U.S. Corn-Belt Yield Prediction

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Yale University

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Code: `https://github.com/petkovip/crop-yields--satellite-image-representation` | Video: `https://youtu.be/G777uqAYG7Q`

Why this matters

- **Crop-yield monitoring** drives agricultural planning, food security, monetary-policy inputs, and is now a recognised input to quantitative finance.
- **Sentinel-2 imagery is free & global**: the bottleneck is representation, not data acquisition.
- **Labels are scarce**: only ∼3 k USDA county-year corn-yield labels across our 7-state Corn Belt sample, vs. ∼850 k unlabelled tiles.
- **Recipe is broadly applicable**: the same gap exists in climate-change attribution, deforestation, biodiversity, urban-growth, and drought monitoring.

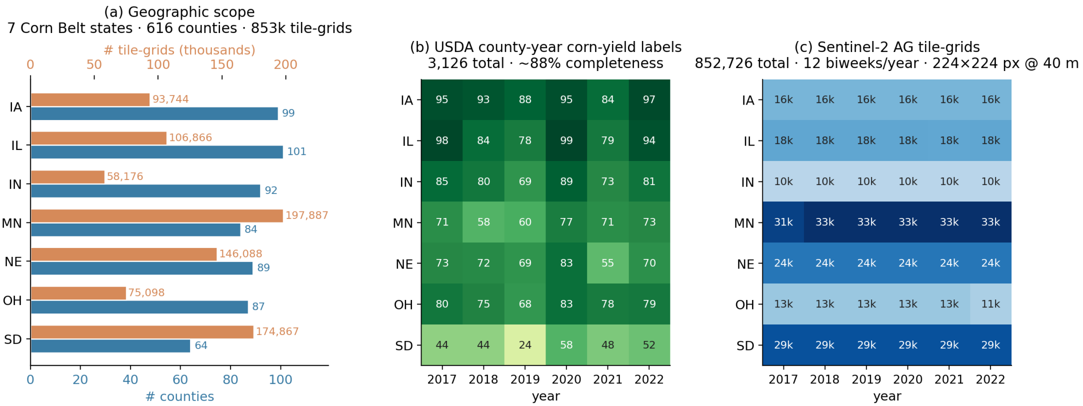
Goal

Learn a rich, transferable representation of multispectral satellite tiles *from unlabelled imagery alone*, then probe it cheaply for yield.

Three eras of crop-yield prediction from satellite imagery

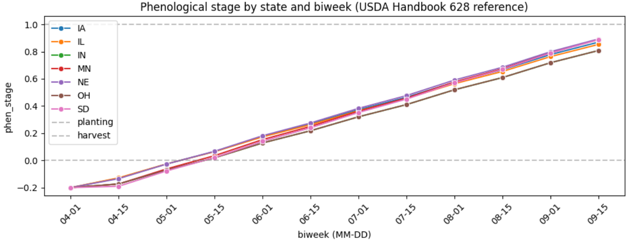
Era	Approach	Examples
Hand-crafted features	Spectral indices $\rightarrow$ tree / linear regressor	NDVI + LightGBM, Khaki et al. 2020 (CNN-RNN, tabular)
Supervised CNN/RNN	End-to-end deep imagery models	ConvLSTM (Shi 2015), CNN-RNN, GNN-RNN (Fan 2022), MMST-ViT (Lin 2024, multi-modal)
Self-supervised	Contrastive / MAE pre-training, then probe	SimCLR (Chen 2020), DINOv2, MAE / Sat-MAE, Prithvi-EO 2.0, SeCo, GASSL
Our work	SimCLR with temporal positives	Geo-SimCLR (this project)

Data: 7 Corn-Belt states, 6 years, biweekly Sentinel-2

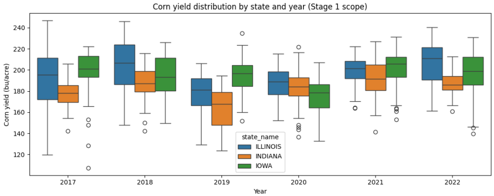


CropNet release of Sentinel-2 imagery – agriculture false-colour bands B02 / B08 / B11 – biweekly Apr–Sep, 224×224 px @ 40 m/pixel (9×9 km tiles).

Phenology window and yield variance

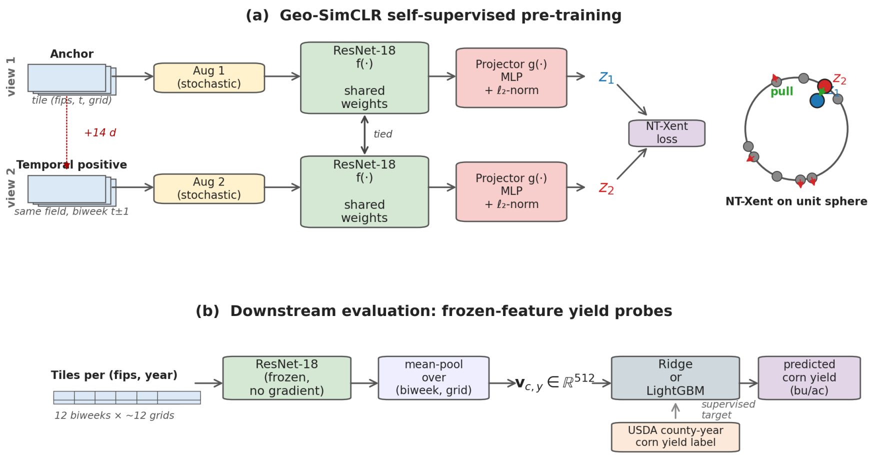


USDA Handbook 628 reference phenology – 7-state coverage of the full April–September growing window.

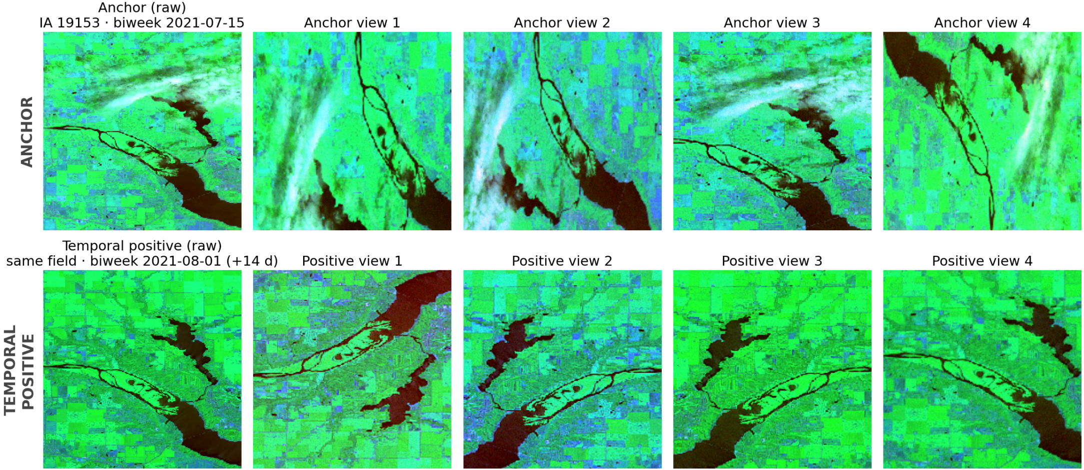


Per-state yield boxplots (early 3-state snapshot; full 7-state numbers in the report).

Geo-SimCLR pipeline



What the contrastive task looks like on real tiles



NT-Xent loss and key training settings

NT-Xent loss (Chen et al. 2020)

$$\mathcal{L}_{i\to j} = -\log \frac{\exp(z_i^\top z_j/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k\neq i]} \exp(z_i^\top z_k/\tau)}$$

ℓ_2-normalised projections z_i, in-batch negatives, temperature $\tau = 0.20$.

Architecture

- ResNet-18 from scratch, 11.2 M params
- Projector MLP 512 $\to$ 256 $\to$ 128 (dropped after SSL)

Optimisation

- AdamW, lr 10^{-3}, wd 10^{-4}
- 5-ep warm-up $\to$ cosine, 50 epochs
- Batch 1024, AMP, $\sim$5 h on 2 GPUs

Pretext-task design

- Positive = same field, biweek $t \pm 1$ ($\sim$14 d)
- CRD-balanced anchors (61 unique CRDs)
- $\sim$645 k anchor pairs / epoch

Augmentations (per view, indep.)

- h/v flip, rot90, crop 160–208 $\to$ 224
- sharpness, intensity jitter, Gaussian noise
- No spectral dropout, RGB jitter, mixup

Evaluation methodology

Splits

- **Time split:** train 2017–2020 / val 2021 / test 2022. Strict no-look-ahead in-season-forecast benchmark.
- **County split:** random 20 % of FIPS held out across all years. Spatial generalisation test.

Probes on frozen features

- Mean-pool tile features per (*fips*, *year*) into one 512-d vector.
- Probe: **Ridge** ($\alpha = 1$) or **LightGBM** (300 trees, early stopping on 2021).
- Optional state one-hot + year scalar: only on county split (year extrapolates badly out-of-sample).
- Bootstrap 95 % CIs from $n = 200$ paired resamples.

Why split-aware probes

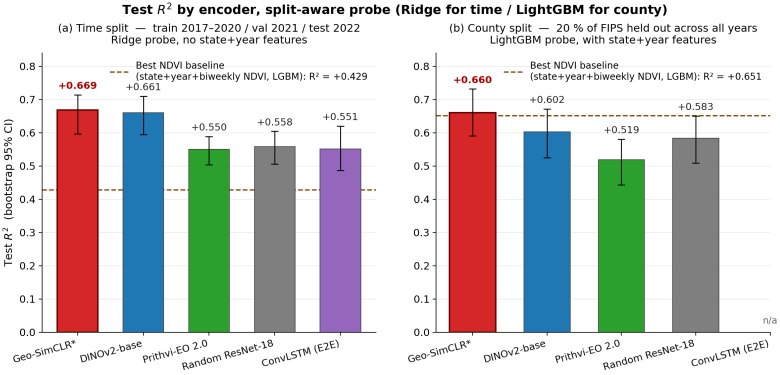
For *every* encoder, Ridge wins time-split and LightGBM with state+year wins county-split. Reported headline cells are the empirical winner per encoder (full grid in the report's Appendix).

Baselines: imagery-only, fixed input modality

- **Hand-crafted NDVI** (6 specifications) $\rightarrow$ Ridge / LightGBM. Strongest: state + year + biweekly NDVI series (LightGBM, 20 features).
- **Random ResNet-18**: same architecture as Geo-SimCLR, never trained. Sanity baseline – mean-pooled random conv features carry colour / texture statistics correlated with vegetation.
- **DINOv2-base** (frozen, 85.7 M params, 768-d). Trained on 142 M natural RGB images; AG bands fed as RGB slots.
- **Prithvi-EO 2.0** (frozen, 303.9 M params, 1024-d). EO-specialised but expects 6-band HLS; we channel-pad the 3 missing.
- **ConvLSTM** end-to-end (12.4 M params). Per-frame ResNet-18 + LSTM(512$\rightarrow$256) + MLP head, z-scored target, image augmentation across all 12 biweek frames.

We deliberately do not chase MMST-ViT: it is multi-modal and the CropNet ablation attributes most of its gain to the WRF-HRRR weather modality, not the imagery.

Headline result: best single encoder on both splits



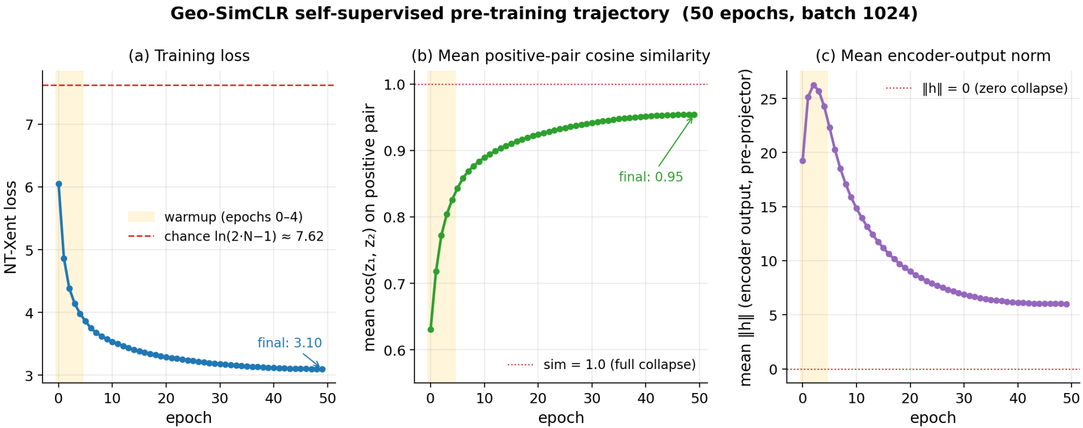
Headline numbers

Geo-SimCLR test $R^2 = \mathbf{0.669}$ (time, Ridge) and **0.660** (county, LightGBM); $+0.118\,R^2$ over supervised end-to-end ConvLSTM on time.

Reading the ranking

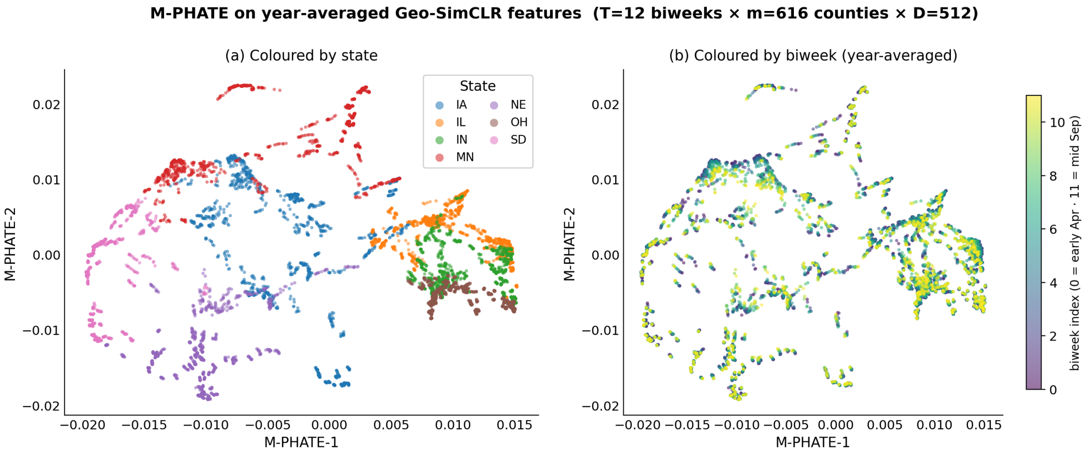
- **Geo-SimCLR $\succ$ ConvLSTM end-to-end** ($+0.118\,R^2$ time) – SSL pretext sees $\sim$6.4 M (anchor, positive, 1023-negs) tuples $\times$ 50 epochs over 850 k unlabelled tiles vs. ConvLSTM's $\sim$2,100 supervised yield labels. Pure data-efficiency win.
- **Geo-SimCLR $\succ$ Prithvi-EO 2.0** (despite 27$\times$ more parameters) – Prithvi expects 6-band HLS; we channel-pad the missing 3. Effective rank on padded input collapses to $\sim$1.7/1024 – documented baseline limitation, not a claim about Prithvi-the-model.
- **Geo-SimCLR $\approx$ DINOv2-base** (within $\sim$0.06 R^2 at 7.6$\times$ fewer parameters) – large general-purpose SSL transfers surprisingly well to satellite RGB-like inputs. Geo-SimCLR's edge comes from the temporal-positive pretext task being domain-aware.

SSL training trajectory (and the first warning sign)



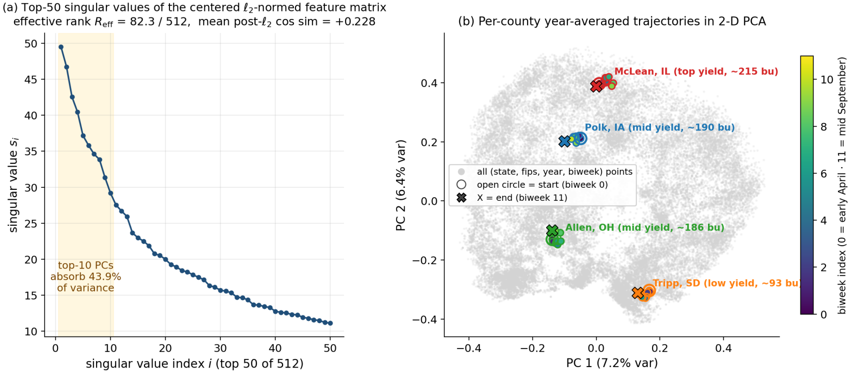
Loss converges cleanly; encoder norm is stable (no zero-collapse). But mean positive-pair cosine similarity climbs to ∼0.95 – adjacent-biweek positives are intrinsically very similar.

What did the encoder learn? – M-PHATE



State coloring (a): clean state-coherent clusters – geography is encoded. **Biweek coloring (b):** smooth but tightly overlapping – seasonal phenology only weakly encoded.

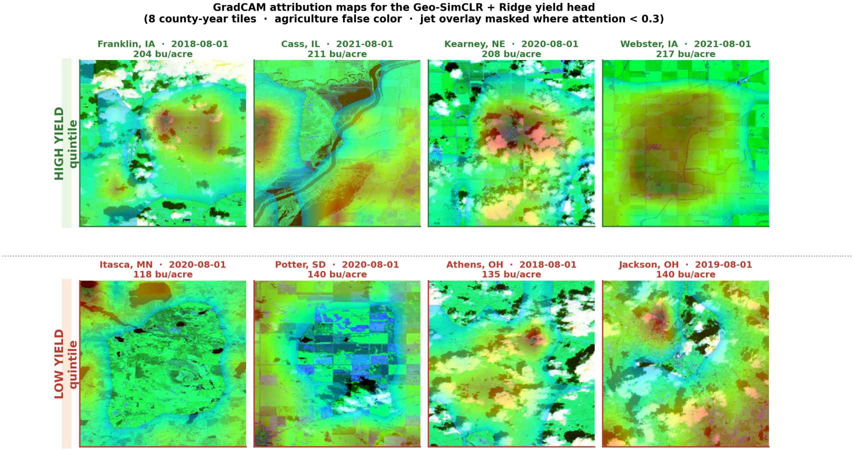
Honest collapse diagnostic



Three numbers that quantify it

Effective rank **82.3 / 512** | mean post-ℓ_2 cosine sim **+0.228** | each county's 12 biweek points span only **4–6 %** of the global pairwise distance.

Pixel attribution via GradCAM



Read in light of the collapse: encoder finds *where* the cropland is – consistent across high- and low-yield extremes – not *how that crop is currently doing*.

Limitations and future work

Limitations

- Partial dimensional collapse along county-identity directions ($R_{\text{eff}} = 82/512$).
- Phenology only weakly encoded – our headline claim is *strong geography*, not *strong phenology*.
- Small backbone (ResNet-18, 11 M params) by SSL-paper standards.
- Evaluation restricted to 7 states / 6 years / corn only.

Future work (priority order)

1. Harder positives + stronger augmentations ($\pm$3–5 biweeks; tighter crop ranges).
2. Decorrelation regulariser (Barlow-Twins / VICReg term added to NT-Xent).
3. Head-to-head against SeCo, GASSL, GeoSSL, SatMAE under the same probe protocol.
4. 4-channel AG+NDVI input; partial fine-tuning; fine-tune on USDA Weekly Crop Progress.
5. Larger backbone, more crops (soybean transfer), weather ablation, cross-country generalisation.

Thank you Questions?

Full report: accompanying NeurIPS-style paper (this folder).
Code: `https://github.com/petkovip/crop-yields--satellite-image-representation`
Video: `https://youtu.be/G777uqAYG7Q`

Selected references

- Chen et al. *A Simple Framework for Contrastive Learning of Visual Representations.* ICML 2020. (SimCLR)
- Lin et al. *An Open and Large-Scale Dataset for Multi-Modal Climate Change-Aware Crop Yield Predictions.* KDD 2024. (CropNet)
- Oquab et al. *DINOv2: Learning Robust Visual Features without Supervision.* TMLR 2024.
- Jakubik et al. *Prithvi-EO-2.0: A Versatile Multi-Temporal Foundation Model for Earth Observation.* IBM-NASA 2024.
- Shi et al. *Convolutional LSTM Network: A Machine Learning Approach for Precipitation Nowcasting.* NeurIPS 2015.
- Gigante et al. *Visualizing the PHATE of Neural Networks.* NeurIPS 2019. (M-PHATE)
- Jing et al. *Understanding Dimensional Collapse in Contrastive Self-Supervised Learning.* ICLR 2022.
- Selvaraju et al. *Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization.* ICCV 2017.